Foundations of Logistic regression

[Logistic regression]

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y=1 \mid x) = \sigma(w^T x + b)$$

1.0 Content Block

where t is the log-odds and β i {\displaystyle \beta _{i}} are parameters of the model. An additional generalization has been introduced in which the base of the model (b) is not restricted to Euler's number e. In most applications, the base b {\displaystyle b} of the logarithm is usually taken to be e. However, in some cases it can be easier to communicate results by working in base 2 or base 10. For a more compact notation, we will specify the explanatory variables and the β coefficients as ■ (M + 1) {\displaystyle (M+1)} ■-dimensional vectors:

2.0 Content Block

Pr (Y i = 1) = e β 1 · X i 1 + e β 1 · X i = 1 1 + e − β 1 · X i = p i {\displaystyle \Pr(Y_{i}=1)={\frac {e^{{\boldsymbol {\beta }}_{1}\cdot \mathbf {X} _{i}}}{1+e^{{\boldsymbol {\beta }}_{1}\cdot \mathbf {X} _{i}}}}={\frac {1}{1+e^{-{\boldsymbol {\beta }}_{1}\cdot \mathbf {X} _{i}}}}=p_{i}}

3.0 Content Block

■ = ∑ k = 1 K y k log b ■ (p (x k)) + ∑ k = 1 K (1 − y k) log b ■ (1 − p (x k)) {\displaystyle \ell =\sum _{k=1}^{K}y_{k}\log _{b}(p({\boldsymbol {x_{k}}}))+\sum _{k=1}^{K}(1-y_{k})\log _{b}(1-p({\boldsymbol {x_{k}}}))}

4.0 Content Block

w k + 1 = (X T S k X) − 1 X T (S k X w k + y − μ k) {\displaystyle \mathbf {w} _{k+1}=\left(\mathbf {X} ^{T}\mathbf {S} _{k}\mathbf {X} \right)^{-1}\mathbf {X} ^{T}\left(\mathbf {S} _{k}\mathbf {X} \mathbf {w} _{k}+\mathbf {y} -\mathbf {\boldsymbol {\mu }} _{k}\right)}

5.0 Content Block

Wald statistic Alternatively, when assessing the contribution of individual predictors in a given model, one may examine the significance of the Wald statistic. The Wald statistic, analogous to the t-test in linear regression, is used to assess the significance of coefficients. The Wald statistic is the ratio of the square of the regression coefficient to the square of the standard error of the coefficient and is asymptotically distributed as a chi-square distribution.

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∑ x ∈ X ∑ y ∈ Y Pr (X = x , Y = y) (− log ■ Pr (Y = y ■ X = x) Pr (Y = y ■ X = x ; θ) + log ■ Pr (Y = y ■ X = x)) =

7.0 Content Block

0 = ∂ ■ ∂ β 1 = ∑ k = 1 K (y k − p k) x k {\displaystyle 0={\frac {\partial \ell }{\partial \beta _{1}}}=\sum _{k=1}^{K}(y_{k}-p_{k})x_{k}}

8.0 Content Block

Widely used, the "one in ten rule", states that logistic regression models give stable values for the explanatory variables if based on a minimum of about 10 events per explanatory variable (EPV); where event denotes the cases belonging to the less frequent category in the dependent variable. Thus a study designed to use k {\displaystyle k} explanatory variables for an event (e.g. myocardial infarction) expected to occur in a proportion p {\displaystyle p} of participants in the study will require a total of 10 k / p {\displaystyle 10k/p} participants. However, there is considerable debate about the reliability of this rule, which is based on simulation studies and lacks a secure theoretical underpinning. According to some authors the rule is overly conservative in some circumstances, with the authors stating, "If we (somewhat subjectively) regard confidence interval coverage less than 93 percent, type I error greater than 7 percent, or relative bias greater than 15 percent as problematic, our results indicate that problems are fairly frequent with 2–4 EPV, uncommon with 5–9 EPV, and still observed with 10–16 EPV. The worst instances of each problem were not severe with 5–9 EPV and usually comparable to those with 10–16 EPV". Others have found results that are not consistent with the above, using different criteria. A useful criterion is whether the fitted model will be expected to achieve the same predictive discrimination in a new sample as it appeared to achieve in the model development sample. For that criterion, 20 events per candidate variable may be required. Also, one can argue that 96 observations are needed only to estimate the model's intercept precisely enough that the margin of error in predicted probabilities is ±0.1 with a 0.95 confidence level.

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t = log b ■ p 1 − p = β 0 + β 1 x 1 + β 2 x 2 + ■ + β M x M {\displaystyle t=\log _{b}{\frac {p}{1-p}}=\beta _{0}+\beta _{1}x_{1}+\beta _{2}x_{2}+\cdots +\beta _{M}x_{M}}

10.0 Content Block

and the maximization procedure can be accomplished by solving the above two equations for ■ β 0 {\displaystyle \beta _{0}} ■ and ■ β 1 {\displaystyle \beta _{1}} ■, which, again, will generally require the use of numerical methods. The values of ■ β 0 {\displaystyle \beta _{0}} ■ and ■ β 1 {\displaystyle \beta _{1}} ■ which maximize ■ and L using the above data are found to be:

11.0 Content Block

Comparison with linear regression Logistic regression can be seen as a special case of the generalized linear model and thus analogous to linear regression. The model of logistic regression, however, is based on quite different assumptions (about the relationship between the dependent and independent variables) from those of linear regression. In particular, the key differences between these two models can be seen in the following two features of logistic regression. First, the conditional distribution y ■ x {\displaystyle y\mid x} is a Bernoulli distribution rather than a Gaussian distribution, because the dependent variable is binary. Second, the predicted values are probabilities and are therefore restricted to (0,1) through the logistic distribution function because logistic regression predicts the probability of particular outcomes rather than the outcomes themselves.

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■ = ∑ k = 1 K ∑ n = 0 N Δ (n , y k) ln ■ (p n (x k)) {\displaystyle \ell =\sum _{k=1}^{K}\sum _{n=0}^{N}\Delta (n,y_{k})\,\ln(p_{n}({\boldsymbol {x}}_{k}))}

13.0 Content Block

Deviance and likelihood ratio tests In linear regression analysis, one is concerned with partitioning variance via the sum of squares calculations – variance in the criterion is essentially divided into variance accounted for by the predictors and residual variance. In logistic regression analysis, deviance is used in lieu of a sum of squares calculations. Deviance is analogous to the sum of squares calculations in linear regression and is a measure of the lack of fit to the data in a logistic regression model. When a "saturated" model is available (a model with a theoretically perfect fit), deviance is calculated by comparing a given model with the saturated model. This computation gives the likelihood-ratio test: